

ELEG418 – Course project

PID controller design for traffic lights
Spring 2024

The purpose of this project is to provide a comprehensive understanding of PID control systems, which are essential in the design and operation of complex systems in various fields. PID controllers are widely used in industrial and manufacturing processes, as well as in robotics, aerospace, and automotive systems, due to their robustness, versatility, and ability to provide accurate and stable control of complex systems.

1 Background

Control systems play a crucial role in many fields, including aerospace, robotics, manufacturing, and transportation. These systems are designed to achieve specific objectives while accounting for various uncertainties and disturbances in the system's environment. Control systems engineering has a wide range of applications, from simple household appliances to complex robotic systems.

1.1 PID control systems

A PID (proportional-integral-derivative) control system is a type of feedback control system used in a wide range of engineering applications. The basic idea behind PID control is to adjust the system inputs based on the difference between the desired output and the actual output, to reduce the error and maintain the desired output.

These PID controllers are used in various systems, including industrial processes, robotics, and automation systems. It is a simple yet effective control technique that can be easily implemented in hardware or software. A PID controller consists of three main components: the *proportional* component, the *integral* component, and the *derivative* component.

- **Proportional Component.** The proportional component of a PID controller adjusts the input to the system in proportion to the error between the desired output and the actual output. The proportional gain parameter, denoted by K_p , determines how much the input is adjusted for a given error. A higher proportional gain results in a larger input adjustment for a given error.
- **Integral Component.** The integral component of a PID controller adjusts the input based on the accumulated error over time. This helps to reduce steady-state error, which is the difference between the desired output and the actual output when the system has reached a steady state. The integral gain parameter, denoted by K_i , determines how much the input is adjusted based on the accumulated error. A higher integral gain results in a larger input adjustment for a given accumulated error.
- **Derivative Component.** The derivative component of a PID controller adjusts the input based on the rate of change of the error. This helps to reduce overshoot and improve the response time of the system. The derivative gain parameter, denoted by K_d , determines how

much the input is adjusted based on the rate of change of the error. A higher derivative gain results in a larger input adjustment for a given rate of change of the error.

1.1.1 PID Control Algorithm

The PID control algorithm combines the three components of the PID controller to adjust the input to the system. The input adjustment is given by the following equation:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (1)$$

where $e(t)$ is the error between the desired output and the actual output at time t .

1.1.2 Advantages and Disadvantages

PID control systems have several advantages, including simplicity, robustness, and adaptability to various applications. However, they also have some limitations, such as difficulty in tuning the parameters and sensitivity to changes in the system dynamics.

2 Traffic light control system

Traffic congestion is a prevalent issue in urban areas worldwide, leading to increased travel times, fuel consumption, and environmental pollution. Effective traffic management systems play a crucial role in mitigating congestion and improving overall transportation efficiency in vehicle intersections [1, 2]. A simple example of a traffic light system in a vehicle intersection is depicted in Figure 1.

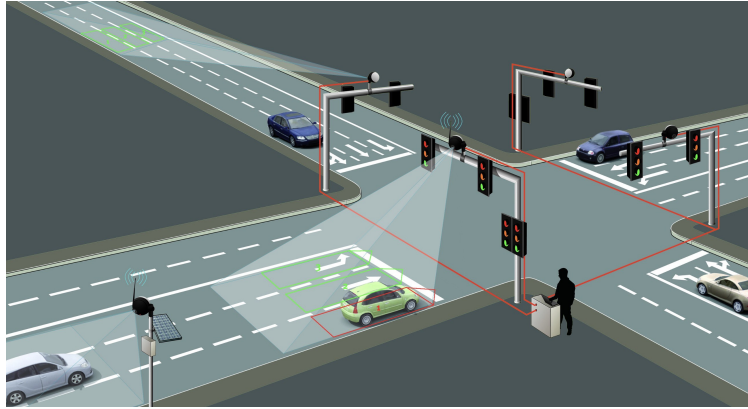


Figure 1: A four-way vehicle intersection with traffic lights.

One of the key components of such systems is traffic signal control, where the timing of traffic lights is optimized to facilitate smooth traffic flow and minimize delays. Traditional traffic signal control systems often operate on fixed timing schedules or basic sensor inputs, which may not adapt well to fluctuating traffic conditions. This can result in inefficient use of road capacity and increased congestion during peak hours [3].

Advanced control strategies, such as Proportional-Integral-Derivative (PID) control, offer a more dynamic and responsive approach to traffic signal control. PID controllers adjust traffic light timings based on real-time traffic data, allowing for better adaptation to changing traffic patterns and improved overall system performance. The key components of a PID-controlled traffic light system include:

- **Plant/manipulator model:** A mathematical model that represents the traffic flow dynamics at the intersection, considering factors such as vehicle arrivals, departures, and queue lengths.
- **Sensors:** Devices that measure real-time traffic conditions, such as inductive loops or cameras, to provide feedback to the controller.
- **PID Controller:** The controller that calculates the necessary adjustments to the traffic light timing based on the error between the desired and actual traffic flow rates.
- **Traffic Light Actuators:** The mechanisms that change the traffic light signals based on the controller's output.

By incorporating a PID controller, the traffic light system can continuously monitor and respond to changing traffic conditions, minimizing congestion and improving overall traffic flow efficiency, as shown in Refs. [1, 2, 4].

3 PID control system design

In this project, we will design a simple PID controller in MATLAB/Python for a traffic light control system at a four-way vehicle intersection. The objective is to regulate the traffic flow efficiently by adjusting the timing of the traffic lights based on real-time traffic conditions. The system should meet the following performance requirements:

- **Settling Time:** The system should reach and maintain the desired traffic flow rate within 3 seconds after a disturbance or change in traffic conditions.
- **Percentage Overshoot (P.O.):** The overshoot in the traffic flow rate should not exceed 5% of the desired steady-state value.

The PID controller should be tuned to achieve the specified settling time and overshoot requirements while minimizing steady-state error. The controller should be robust enough to handle variations in traffic patterns and external disturbances, such as accidents or road closures.

Read the instructions in the following subsections to develop a MATLAB/Python *simulation* of the traffic light control system, including the PID controller and appropriate plant model(s). Clearly document your design process, including the selection of controller parameters, simulation results, and any assumptions or simplifications made. Upon completion of the project, you will be required to submit a written report describing your work, along with the MATLAB/Python code and *simulation* results.

Instructions. Create a concise report based on your procedure and solutions to the questions asked. Submit your report as a single PDF file, and all the .m or .py files you create for the 10% of your course grade. Use the following MS Word or LaTeX templates to create your report.

3.1 Manipulator/plant for a traffic light

Our problem is restricted to a particular vehicle intersection where a study has been conducted to estimate the dynamics of traffic flow at the intersection by considering five different variables:

- *Vehicle arrivals:* The rate at which vehicles arrive at the intersection from each direction.
- *Vehicle departures:* The rate at which vehicles exit the intersection after receiving a green light.
- *Queue dynamics:* The formation and dissipation of vehicle queues at each approach to the intersection.
- *Intersection geometry:* The number of lanes, turn lanes, and other physical characteristics of the intersection.
- *Traffic signal timing:* The timing and sequence of the traffic light phases (green, yellow, red) for each approach.

From the study, a mathematical model that represents the dynamics of traffic flow at the intersection has been derived. The system describing this characteristic is modeled using the transfer function $G(s) = \frac{3}{(s+0.1)(s+0.5)}$, as shown in Figure 2.

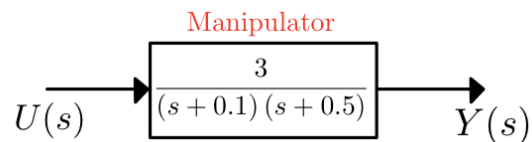


Figure 2: Transfer function describing the traffic flow at the intersection.

As a first inspection, we can start analyzing the manipulator (plant) of the system using *computer tools*. Proceed and answer the following questions employing MATLAB/Python.

1. Determine the step response of the manipulator (open-loop response).
2. Based on the graph, what conclusion can you draw? Is the system stable? If so, what type of system is this? (underdamped, overdamped, etc.)
3. Plot the bode diagram response for the manipulator (magnitude and phase). At what frequency ω does the system's gain drop to 70.7% (i.e., exhibit a decay of -3 dB)?
4. What changes would be required in the plant's transfer function to increase this cut-off frequency?

5. A feedback loop is added to the manipulator to consider the steady-state errors. Obtain the step response for the *unit* closed-loop transfer function of the plant alone. What differences can you identify compared to the open-loop step response?

As seen from the previous experiments, although the system is stable, it takes a considerable amount of time for the closed-loop system to stabilize. This is not ideal for operating the traffic lights in the intersection. Therefore, to decrease the system's response, we need to introduce a controller in the open path of the closed-loop system. One alternative to reduce the settling time of system response for the traffic lights is using a PID controller, as shown in Figure 3.

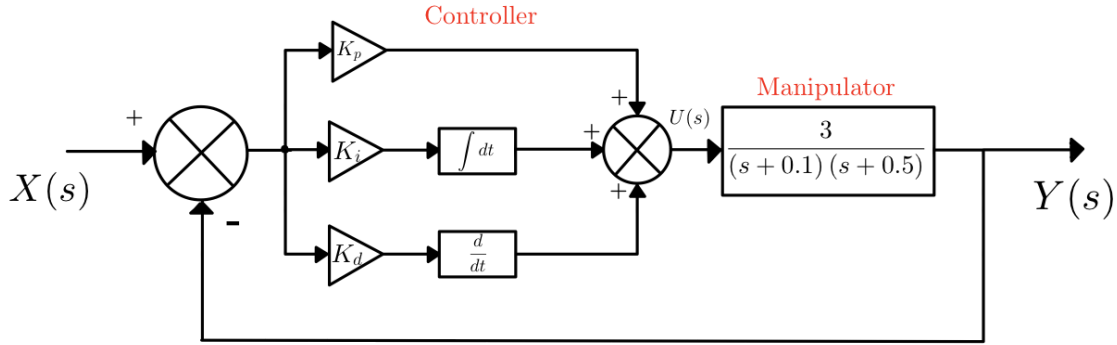


Figure 3: PID controller scheme for a traffic light intersection.

Using the block diagram configuration presented above, the goal is to design a P, I, D, PI, PD, or PID controller in MATLAB/Python (as your preference), utilizing root locus (or any other method) to meet the following requirements:

6. Explain the reasoning behind your choice of controller type.
7. Design a controller to achieve a system with percent overshoot to a step input of $P.O. \leq 5\%$.
8. Determine the gain(s) that minimizes the settling time (with a 2% criterion) while maintaining a percent overshoot of $P.O. \leq 5\%$.
9. Plot the step response of the system with the designed controller.
10. What conclusions did you obtain after designing and testing the controller for the traffic light system?

In addition to your design, an operator has decided to implement the PID controller shown in Figure 3 as an alternative solution for this problem. For this particular controller, the operator simplifies the overall gain to K , where, $K_p = 2K$, $K_i = 8K$, and $K_d = 0.5K$.

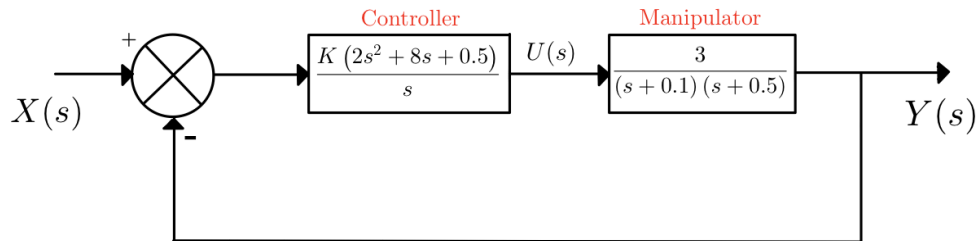


Figure 4: PID controller applied to the traffic intersection's plant.

To meet the same requirements as in the previous design, use MATLAB/Python to determine the appropriate value of K and answer the following questions:

11. Inspect the root locus plot and identify the roots for $K = 2$ and 10. What is the damping ratio and P.O. for each value of K ?
12. Determine the gain K that results in a percent overshoot to a step of $P.O. \leq 5\%$.
13. Determine the gain that minimizes the settling time (with a 2% criterion) while maintaining a percent overshoot of $P.O. \leq 5\%$.
14. Plot the step and impulse response of the system with the value of K found.
15. What conclusions can you draw from this analysis?
16. How does this controller compare with your design? Which of the two controllers would you choose to control the traffic lights?

References

- [1] N. Fairfield and C. Urmson, "Traffic light mapping and detection," in *2011 IEEE International Conference on Robotics and Automation*, pp. 5421–5426, IEEE, 2011.
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